

E-Commerce Customer Analytics Report

Olist Brazilian E-Commerce | 2016 to 2018

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1. Project Background

This project analyses 100,000 real orders from Olist, a Brazilian e-commerce marketplace, covering the period from September 2016 to August 2018. The goal was to understand revenue trends, identify customer behaviour patterns, and segment customers to support business decision-making.

The analysis was done across three tools. MySQL was used for data cleaning and aggregation, Python for RFM scoring and K-Means clustering, and Power BI for the final interactive dashboard.

2. Dataset Overview

Table	Rows
Orders	99,441
Order Items	118,291
Customers	99,441
Payments	103,886
Products	32,340
Seller	3,095

After cleaning, 96,470 delivered orders were kept for analysis. Cancelled, unavailable, and orders with missing dates were removed.

3. Key Findings

Revenue

Total revenue across the period was R\$13.4 million from 92,850 unique orders. Revenue grew consistently from late 2016 through mid-2018, with a sharp dip in September 2017 followed by recovery. The average order value was R\$120.

Top Categories

Beauty and health were the highest revenue category, followed by watches and gifts, bed and bath, and sports and leisure. These four categories alone account for a huge part of total revenue and should be prioritised in marketing spend.

Payment Behaviour

73.9% of all transactions were paid by credit card. Boleto (a Brazilian payment slip) was second at 19%. Credit card customers also used instalments on average 3.5 times per transaction, meaning customers are splitting large purchases. This

is an opportunity to offer installment-based promotions to increase average order value.

Customer Behaviour

Every single customer in this dataset placed exactly one order. The repeat purchase rate is effectively zero. This is the biggest business problem in the entire dataset. Acquiring a customer costs money. If they never come back, the business is constantly spending on acquisition with no return on retention.

4. RFM Customer Segmentation

Customers were scored on three dimensions: how recently they bought (Recency), how many times they bought (Frequency), and how much they spent (Monetary). Based on combined scores, 92,850 customers were placed into five segments.

Segment	Customers	Avg. Revenue
Champions	7,497	₹296
Loyal	31,756	₹193
At risk	38,344	₹112
Hibernating	14,458	₹50
Lost	795	₹50

Champions spend 11 times more than Lost customers. But they represent only 8% of the customer base. The At-Risk segment is the most critical one. It is the largest group at 41.3% of all customers and they are drifting away. If even 20% of At-Risk customers can be re-engaged, that is significant revenue recovery.

K-Means clustering with k=4 confirmed these behavioural patterns, identifying four distinct groups based on recency and monetary value.

5. Business Recommendations

Re-engage At Risk customers immediately

38,344 customers have not purchased in an average of 268 days. A targeted email or discount campaign specifically for this segment could recover a meaningful portion of revenue. Even a 10% conversion rate brings back 3,800 customers.

Build a loyalty program

Since every customer currently buys only once, there is no structural reason for them to return. A points-based loyalty program or cash-back offer tied to repeat purchases directly address the zero-retention problem.

Double down on Beauty and Electronics

These categories drive the most revenue. Allocating more marketing budget here, and expanding seller inventory in these categories, will have the highest return.

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Offer instalment incentives

Credit card customers already use instalments 3.5 times on average. Promoting higher-value products with zero-interest instalment plans could increase average order value further.

Focus on Champions

These 7,497 customers spend R\$296 on average and bought recently. They are the most likely to respond to premium offers, early access to new products, or a VIP program. Keeping them engaged is cheaper than acquiring new customers.

6. Methodology Summary

1. Downloaded Olist Brazilian E-Commerce dataset from Kaggle (9 CSV files)
2. Imported into MySQL and cleaned data, keeping only 96,470 delivered orders

3. Built aggregation tables for monthly revenue, category performance, payment analysis, and seller performance
4. Exported cleaned data to Google Colab for Python analysis
5. Performed RFM scoring using quintile-based ranking across all 92,850 customers
6. Applied K-Means clustering (k=4) using Recency and Monetary as features
7. Built a 4-page interactive Power BI dashboard covering business overview, customer segmentation, payment analysis, and RFM deep dive